Chapter 7 Estimating Parameters and Determining Sample Sizes
Estimating a Population Proportion (7.1)

Objectives

In this lesson, we will learn how to

- use a sample proportion as a point estimate of a population proportion;
- construct and interpret a confidence interval for a population proportion;
- find the margin of error for a confidence interval;
- determine a required sample size for estimating a population proportion;
- understand why some confidence interval methods perform better than others.

1 Point Estimate for a Population Proportion

A point estimate is a single value used to estimate a population parameter.

For a population proportion, the best point estimate is the sample proportion.

$$\hat{p} = \frac{x}{n}$$

where

- x is the number of successes in the sample;
- n is the sample size;

- $\hat{p}$ is read as ``p-hat.''

Point Estimate of p

The sample proportion $\hat{p}$ is the best point estimate of the population proportion p.

Example 1: Online Courses: A survey of 950 undergraduate college students found that 53% of them said they take online courses. Find the best point estimate of the proportion of all undergraduate college students who take online courses.

Solution

Because the sample proportion is the best point estimate of the population proportion,

$$\hat{p} = 0.53.$$

So the best point estimate of the population proportion is

$$\boxed{0.53}$$

or 53%.

2 Confidence Intervals

A confidence interval is an interval of values used to estimate the true value of a population parameter.

Confidence Interval

A confidence interval gives a range of reasonable values for the population parameter.
For example,

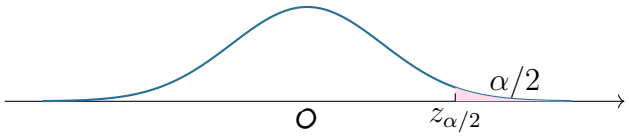
$$0.405 < p < 0.455$$

means that the interval from 0.405 to 0.455 is used to estimate the true population proportion p.

The confidence level is the success rate of the method in the long run. Common confidence levels are:

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.96
99%	0.01	2.575

Standard normal critical value



Correct Interpretation

For a 95% confidence interval, we say:

We are 95% confident that the interval contains the true population proportion p.

We do not say that there is a 95% chance that p falls in the interval after the interval has already been computed.

3 Requirements for a Confidence Interval for p

Requirements

To construct a confidence interval for a population proportion p, check that:

1. the sample is a simple random sample;
2. the conditions for a binomial experiment are satisfied;

3. there are at least 5 successes and at least 5 failures.

For the third requirement, check:

$$x \geq 5 \qquad \text{and} \qquad n - x \geq 5.$$

4 Margin of Error and Confidence Interval

The margin of error, denoted by E, is the maximum likely amount of error when $\hat{p}$ is used to estimate p.

Margin of Error for a Population Proportion

$$E = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

where

$$\hat{q} = 1 - \hat{p}.$$

Confidence Interval for a Population Proportion

$$\hat{p} - E < p < \hat{p} + E$$

or

$$\hat{p} \pm E.$$

Round-Off Rule

Round confidence interval limits for p to three significant digits.

5 Constructing a Confidence Interval

Example 2: Online Courses Confidence Interval: In a survey of 950 undergraduate students, 53% said they take online courses.

(a) Find the margin of error for a 95% confidence interval.

(b) Find the 95% confidence interval estimate of the population proportion p.

(c) Can we safely conclude that more than 50% of undergraduates take online courses?

Solution

We are given

$$n = 950, \qquad \hat{p} = 0.53, \qquad \hat{q} = 1 - 0.53 = 0.47.$$

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

(a) Margin of error:

$$E = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

$$E = 1.96\sqrt{\frac{(0.53)(0.47)}{950}}$$

$$E \approx 0.0317.$$

(b) Confidence interval:

$$\hat{p} - E < p < \hat{p} + E$$

$$0.53 - 0.0317 < p < 0.53 + 0.0317$$

$$0.4983 < p < 0.5617.$$

Rounded to three significant digits,

$$\boxed{0.498 < p < 0.562}.$$

(c) Conclusion: Because the interval includes values below 0.50, we cannot safely conclude that more than 50% of all undergraduates take online courses.

Accurate Summary Statement

A good way to summarize the result is:

53% of undergraduates in the sample take online courses. Based on the survey of 950 undergraduates, we are 95% confident that the true population percentage is between about 49.8% and 56.2%.

6 Analyzing Polls

When reading poll results, always check the following:

- Was the sample selected in a reasonable way?
- Was the sample size provided?
- Was the confidence level provided?
- Was the margin of error provided?

Important Polling Idea

For most large populations, the quality of a poll depends mostly on the sampling method and the sample size, not on the size of the population.

7 Finding the Point Estimate and Margin of Error from a Confidence Interval

Sometimes a confidence interval is already given, and we want to recover the point estimate and margin of error.

Point Estimate and Margin of Error from a Confidence Interval

If a confidence interval has lower limit L and upper limit U, then

$$\text{Point estimate} = \frac{L+U}{2}$$

and

$$E = \frac{U-L}{2}.$$

Example 3: Nicotine Patch Therapy: A study reported that 70% of 71 subjects were abstinent from smoking at 8 weeks, with a 95% confidence interval from 58% to 81%.
Use the confidence interval to find the point estimate and the margin of error.

Solution

The confidence interval is

$$0.58 < p < 0.81.$$

The point estimate is the midpoint:

$$\frac{0.58+0.81}{2} = 0.695.$$

The margin of error is half the width:

$$E = \frac{0.81-0.58}{2} = 0.115.$$

Therefore,

$$\boxed{\text{point estimate} = 0.695, \qquad E = 0.115.}$$

8 Sample Size for Estimating a Population Proportion

Before collecting data, we often need to decide how large the sample should be.

Sample Size When an Estimate of $\hat{p}$ Is Known

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{E^2}$$

where

$$\hat{q} = 1 - \hat{p}.$$

Sample Size When No Estimate of $\hat{p}$ Is Known

If no prior estimate of $\hat{p}$ is available, use

$$\hat{p} = 0.5, \qquad \hat{q} = 0.5.$$

Then

$$n = \frac{z_{\alpha/2}^2 (0.25)}{E^2}.$$

Round-Off Rule for Sample Size

If the computed sample size is not a whole number, always round up to the next whole number.

9 Finding a Required Sample Size

Example 4: Online Purchases: A previous Pew Research Center survey found that 79% of U.S. adults shop online. We want to conduct a new survey and be 95% confident that the sample percentage is in error by no more than 3 percentage points.

(a) Assume that 79% of adults make online purchases.

(b) Assume that we have no prior information.

Solution

The margin of error is

$$E = 0.03.$$

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

(a) Using the prior estimate $\hat{p} = 0.79$:

$$\hat{q} = 1 - 0.79 = 0.21.$$

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{E^2}$$

$$n = \frac{(1.96)^2(0.79)(0.21)}{(0.03)^2}$$

$$n \approx 708.1.$$

Round up:

$$\boxed{n = 709.}$$

(b) With no prior information: Use $\hat{p} = 0.5$ and $\hat{q} = 0.5$.

$$n = \frac{(1.96)^2(0.25)}{(0.03)^2}$$

$$n \approx 1067.1.$$

Round up:

$$\boxed{n = 1068.}$$

With no prior estimate, we need a larger sample size.

10 Better Performing Confidence Intervals

The method introduced above is often called the Wald confidence interval. It is useful for teaching and for basic calculations, but it is not always the best-performing method.

Plus Four Method

The plus four method adjusts the data before constructing the confidence interval:

- add 2 successes;
- add 2 failures;
- increase the sample size by 4;
- then construct the confidence interval using the usual method.

This method often performs better than the basic Wald interval.

Other methods include the Wilson score interval and the Clopper-Pearson exact method. These methods can perform better, but they are more complicated and are usually handled with technology.

Summary

Key Ideas

- The sample proportion $\hat{p}$ is the best point estimate of the population proportion p.
- A confidence interval estimates the true population proportion using an interval of values.
- For a population proportion,

$$E = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}.$$

- A confidence interval can be written as

$$\hat{p} - E < p < \hat{p} + E.$$

- To determine sample size, use

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{E^2}.$$

- If no prior estimate is available, use $\hat{p} = 0.5$ and $\hat{q} = 0.5$.
- Always round sample size up.